

Preprocessing and Validation Protocol

A complete, model-facing record of how the advanced draft-entry classifier and early-career minutes regressor turn processed basketball data into fitted, evaluated artifacts—and why each transformation exists.

Draft-entry classifier

Combine participants · 107 predictors · calibrated annual ranking

Minutes regressor

Drafted players · 48 predictors · raw three-year NBA minutes

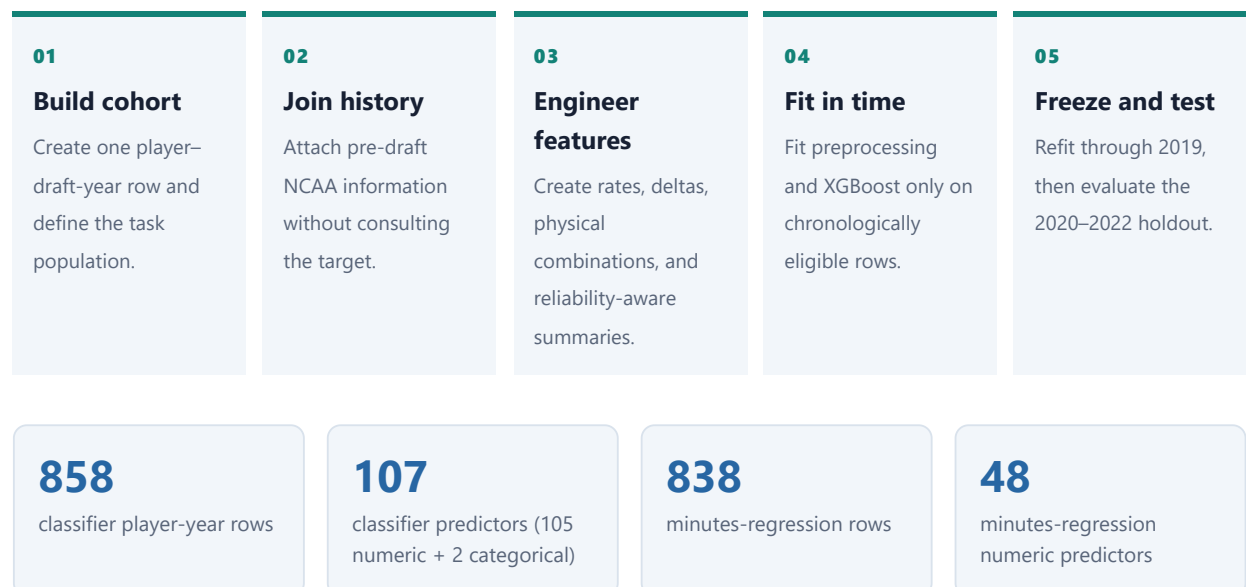
What this report covers

This document records every model-facing preprocessing step used by the restored best configurations: cohort assembly, target-independent linkage, feature construction, fit-time transformations, temporal validation, group-overlap control, probability calibration, and final decision logic.

Scope boundary. This is a modeling-method report, not a catalogue of every HTTP request or source-file normalization inside the separate data-collection package. It starts where raw Combine, NCAA, draft-history, and NBA outcome tables are converted into modeling rows.

Model	Question	Population	Frozen configuration
Draft entry	Will this Combine participant be drafted in the same year?	Players present in all four collected Combine source rosters, 2009–2022	<code>configs/xgboost_draft_classifier_advanced_best.json</code>
NBA minutes	How many regular-season minutes will a drafted player receive in his first three NBA seasons?	Drafted players with a known three-season minutes target, 2009–2022	<code>configs/xgboost_minutes_advanced_best.json</code>

End-to-end flow



Constructing the draft-entry cohort

1. Define a stable Combine population

1. Column names and positions are normalized before the sources are combined.
2. Measurements are converted to numeric values. Invalid numeric text becomes missing rather than a fabricated zero.
3. Rows without a usable player-year key are removed.
4. Duplicate player-year measurements within a source are collapsed: numeric values use their mean; position uses a deterministic mode with a stable tie-break.
5. Four Combine sources are inner-joined, so every retained player belongs to the consistently observed post-Combine cohort.
6. Columns that are globally empty after assembly are dropped. Individual missing measurements remain missing.

Why. Requiring the common four-source roster makes year-to-year membership more comparable. Retaining genuine measurement gaps avoids inventing information and avoids “availability” flags that might mostly reveal the collection era.

2. Define the target without model features

`drafted = 1` only when the normalized NBA player identifier and draft year occur together in official draft history; otherwise it is `0`. Draft pick, round, team, and every post-draft NBA outcome remain audit columns only and are forbidden as predictors.

3. Attach NCAA history without looking at drafted status

Step	Implementation	Reason
Candidate linkage	Exact compact-name matches in the same season; limited qualifier handling for known UNC/UMD ambiguity; otherwise a unique athlete within the prior six seasons.	Keep the linkage deterministic and independent of the outcome.
Ambiguity	Ambiguous or unresolved histories remain missing.	Do not choose a college identity because the player was drafted.
Transfers	Team rows are summed; primary team and position come from the highest-minute row, then games/team provide deterministic tie-breaks.	Represent the player’s full season while retaining one categorical position.
Lookback	Only the draft season and the preceding six seasons are eligible.	Prevent future college information from entering the feature vector.
Age	Age is measured on July 1 of the draft year; values outside 16–35 become missing.	Use a common basketball-age reference and reject implausible source values.

Rates, physical traits, and development signals

All feature engineering is target-blind. It uses only measurements already attached to the prospect, or—in transformations that were considered but not selected—other prospects in the same event year.

Safe rate calculations

Per-40 production
 $40 \times \text{statistic} \div \text{minutes}$
Effective FG%
 $(\text{FGM} + 0.5 \times 3\text{PM}) \div \text{FGA}$
True shooting
 $\text{points} \div [2 \times (\text{FGA} + 0.44 \times \text{FTA})]$
Playmaking rates
 $\text{AST} \div \text{TOV}; 3\text{PA} \div \text{FGA}; \text{FTA} \div \text{FGA}$

Zero denominators, infinities, and undefined rates are changed to missing. No undefined basketball ratio is encoded as zero.

Combine-derived measurements

Feature	Formula	Purpose
Length advantage	wingspan – height without shoes	Separate reach/length from raw height.
Standing reach advantage	standing reach – height without shoes	Represent functional reach.
Maximum reach	standing reach + max vertical	Combine static length and explosiveness.
Vertical improvement	max vertical – standing vertical	Describe the run-up contribution.
BMI	$703 \times \text{weight} \div \text{height}^2$	Compact mass-for-height measure; interpreted cautiously for athletes.

NCAA summaries

Last-season and career values are produced separately. Career totals are aggregated before rates are recomputed. Development features are last-season minus earlier/career values, which captures direction of change rather than only the final level.

Stocks
 $\text{steals per 40} + \text{blocks per 40}$
Playmaking margin
 $\text{assists per 40} - \text{turnovers per 40}$
Production index
 $\text{PTS} + \text{REB} + \text{AST} + 2(\text{STL} + \text{BLK}) - \text{TOV}, \text{all per 40}$
Usage proxy
 $\text{FGA} + 0.44 \times \text{FTA} + \text{TOV}, \text{all per 40}$
Shrunk 3P%
Shrunk FT%

$$(3PM + 0.35 \times 40) \div (3PA + 40)$$

$$(FTM + 0.72 \times 30) \div (FTA + 30)$$

Youth score

23 – age; also age – 21 and age²

Age-adjusted output

youth score × scoring or stocks

Why shrink percentages. A raw shooting percentage based on very few attempts is unstable. The fixed prior adds a modest amount of league-like evidence without learning a statistic from the future folds.

Preprocessing inside the model pipeline

These transformations are fitted anew wherever a model is fitted. Fold-local fitting prevents future validation rows from determining imputation or category levels.

Operation	Exact behavior in the advanced config	Why
Schema validation	Require configured features, target, year, ID/group columns; require one row per <code>(nba_player_id, draft_year)</code> ; target must be binary.	Fail early on a stale or duplicated dataset.
Leakage guard	Reject target, draft pick/round/team, IDs/names, absolute year/season fields, match metadata, and every <code>nba_*</code> outcome feature.	Only information available before the draft decision may enter.
Numeric cleanup	Coerce to numeric; replace $\pm$ infinity with <code>NaN</code> .	Give undefined values one consistent representation.
Numeric missingness	<code>numerical_missing_strategy = native</code> : pass values and <code>NaN</code> directly to XGBoost. No median, standardization, or missing indicator.	XGBoost learns a default missing branch at each split; preserving the gap avoids invented median measurements.
Categorical missingness	Convert pandas <code>pd.NA</code> to ordinary <code>np.nan</code> , then fill with literal <code>missing</code> .	Fix sklearn's ambiguous-null behavior and retain "not observed" as a category.
Categorical encoding	Float32 one-hot encoding; unknown categories are ignored.	A tree receives numeric columns and future unseen position labels do not crash scoring.
Class weighting	The tuned XGBoost parameter <code>scale_pos_weight = 0.3</code> is retained.	The cohort has more positives than negatives; the value was selected in pre-test folds.

Missing does not mean zero. A missing vertical leap, NCAA history, or shooting rate is never converted to a physical measurement of zero. Native tree handling or the explicit categorical sentinel preserves that distinction.

Chronological and entity-safe splits

Period	Years	Rows	Use
Training	2009–2017	510	Base estimation and calibration-model training.
Validation	2018–2019	146	Dedicated sigmoid calibration and annual decision-rule fitting.
Test	2020–2022	202	Final chronological holdout; not used to fit transformations or parameters.

Before calibration, six earlier rows belonging to players who reappear in validation are removed from the training data. Before the final refit, eight train/validation rows belonging to players who reappear in test are removed. The final estimator therefore fits 648 rows.

Hyperparameter tuning without using the holdout

Expanding annual folds

Validation year	Training cutoff	Training rows after purge	Validation rows
2012	≤ 2011	152	61
2013	≤ 2012	213	63
2014	≤ 2013	276	56
2015	≤ 2014	332	58
2016	≤ 2015	390	56
2017	≤ 2016	441	64

Every candidate receives a newly fitted preprocessor and estimator in every fold. The configured search evaluates 180 reproducibly sampled candidates plus two fixed reference candidates, deduplicated if they coincide.

Selection metric

For year y : **normalized AP** = $(AP_y - \text{prevalence}_y) \div (1 - \text{prevalence}_y)$. The candidate score is the unweighted mean across years.

This converts each year's no-skill baseline to zero and a perfect ranking to one. Macro-averaging prevents a large draft class or a high-positive-rate year from dominating model selection. The winning pre-test score was **0.560074**.

Frozen XGBoost parameters

Parameter	Value	Role
<code>n_estimators</code> / <code>learning_rate</code>	400 / 0.035	Many small boosting steps.
<code>max_depth</code> / <code>min_child_weight</code>	1 / 3	Strongly regularized shallow interactions.
<code>subsample</code> / <code>colsample_bytree</code>	0.65 / 0.8	Row and feature sampling for variance control.
<code>reg_alpha</code> / <code>reg_lambda</code>	0 / 3	L1/L2 leaf regularization.
<code>gamma</code> / <code>max_delta_step</code>	0.5 / 1	Conservative new splits and bounded leaf updates.
<code>scale_pos_weight</code>	0.3	Class contribution selected in the same pre-test folds.
Reproducibility	<code>hist</code> , seed 42	Deterministic configuration for the recorded software stack.

From ranking score to reported probability and class

Sigmoid probability calibration

1. Remove six training rows whose player appears in 2018–2019.
2. Fit a separate XGBoost model through 2017 and score 2018–2019.
3. Clip raw probabilities away from exactly 0 and 1, then convert them to log-odds.
4. Fit logistic regression (`C=1`, `lbfgs`, seed 42) from raw log-odds to the validation labels.
5. Freeze this map and apply it to scores from the final estimator.

Fitted map: coefficient **0.448170**, intercept **0.308034**. Calibration changes probability scale but, because it is monotonic, it does not change AP, ROC-AUC, or within-year ranking.

Annual top-fraction rule

The validation positive rate is $92/146 = 0.6301369863$. For every draft year independently, the model:

1. sorts calibrated scores descending with a stable tie-break;
2. selects `floor(0.6301369863 × year_rows + 0.5)` prospects;
3. labels those prospects as predicted drafted.

Operational scope. This rule is valid only when the input contains the complete Combine-participant batch for each scored year. It is not an independent one-row probability threshold.

Final refit and holdout

The final XGBoost estimator is refitted on train + validation after removing eight earlier rows for players who appear in test. The frozen calibrator and annual rule are then applied to 2020–2022.

Holdout metric	Value	Interpretation
Average precision	0.824542	Primary ranking quality; unchanged by sigmoid calibration.
ROC-AUC	0.750620	Pairwise ranking discrimination.
Raw / calibrated log loss	0.593872 / 0.602714	Calibration did not improve this holdout's log loss.
Balanced accuracy	0.657051	Annual-rule class performance.
Precision / recall / F1	0.732283 / 0.750000 / 0.741036	Annual-rule operating metrics.
Confusion matrix	TN 44 · FP 34 · FN 31 · TP 93	202 total test rows.

Evidence boundary. The feature family and XGBoost parameters were frozen before holdout scoring, so AP and ROC-AUC are the clean ranking claims. The annual decision rule was repaired after the first holdout run exposed a refit score-scale mismatch; its thresholded metrics require confirmation on a future cohort. The calibrator was learned from a train-only model but applied to a refitted model, so calibrated probability quality is also not a pristine independent claim.

Advanced early-career minutes preprocessing

The regression task is conditional: it asks about early NBA playing time only after a player has been drafted. Draft capital is therefore deliberately included. This is a different estimand from a strictly pre-draft college-only model.

Rows and target

- Input: `drafted_players_advanced.parquet`, 838 retained rows.
- Target: raw `nba_min_3y`, the sum of regular-season minutes across the first three NBA seasons.
- Rows missing the target or draft year are dropped.
- The configured target is **not** log-transformed.
- Splits: 540 train rows through 2017, 120 validation rows in 2018–2019, and 178 test rows in 2020–2022.

Materialized feature engineering

Family	Transformation	Why
Missing age	Binary indicator for missing draft-year age.	Median imputation otherwise hides that age was unavailable.
College stocks	Last-season steals + blocks per 40, requiring both inputs.	Compact defensive-event signal while preserving partial missingness.
Development differences	Last season – career for seven configured statistics.	Represent recent improvement or decline.
Draft capital	log, square root, inverse, square, top-5/top-10/lottery/first-round flags.	Early opportunity is strongly nonlinear in pick number.
Age and interaction	Centered age, age ² , youth score, pick × centered age.	Allow nonlinear age effects and investment-by-age interaction.
College composites	Shrunk shooting, stocks, playmaking margin, production index, age-adjusted output.	Stabilize small samples and expose basketball-relevant combinations.
Last/career ratios	Last-season value ÷ career value; zero denominators become missing.	Scale-aware development signal.

Fit-time pipeline

1. Validate all configured columns; numeric-coerce target, year, and 48 predictors; replace infinities with missing.
2. Fit median imputation on the training rows for every numeric predictor.
3. Fit standardization (zero mean, unit variance) after imputation.
4. Fit the configured XGBoost regressor. Although trees do not require scaling, the generic regression pipeline applies it consistently.
5. For the frozen best run, refit on train + validation and report 2020–2022 as the holdout.

Reproducibility disclosure. The JSON `feature_engineering` block is provenance; current `train_model.py` does not execute it. The configured derived columns are already materialized in the parquet file. Current `tune_model.py` does replay the configured three expanding validation folds and fixed reference candidate, while direct training uses the frozen winning parameters without starting a search.

Frozen holdout result	Value
Test rows	178 (2020–2022)
R ²	0.481949
RMSE	1,568.183 minutes

APPENDIX A

Advanced classifier feature inventory

The selected `age_college_composites` family contains exactly 105 numeric and two categorical predictors. Relative draft-class percentiles and last-to-career ratio families were generated during exploration but are not in this winner.

Anthropometric measurements · 8

`combine_anthro_height_wo_shoes` · `combine_anthro_height_w_shoes` · `combine_anthro_weight` · `combine_anthro_wingspan` · `combine_anthro_standing_reach` · `combine_anthro_body_fat_pct` · `combine_anthro_hand_length` · `combine_anthro_hand_width`

Athletic drills · 6

`combine_drill_standing_vertical_leap` · `combine_drill_max_vertical_leap` · `combine_drill_lane_agility_time` · `combine_drill_modified_lane_agility_time` · `combine_drill_three_quarter_sprint` · `combine_drill_bench_press`

Combine-derived · 5

`combine_derived_wingspan_minus_height` · `combine_derived_standing_reach_minus_height` · `combine_derived_max_reach` · `combine_derived_max_minus_standing_vertical` · `combine_derived_bmi`

Last NCAA season · 32

`college_last_season_gap` · `college_last_games_played` · `college_last_games_started` · `college_last_minutes` · `college_last_minutes_per_game` · `college_last_start_rate` · `college_last_field_goals_attempted` · `college_last_three_point_field_goals_attempted` · `college_last_free_throws_attempted` · `college_last_fg_pct` · `college_last_three_pct` · `college_last_ft_pct` · `college_last_efg_pct` · `college_last_ts_pct` · `college_last_points_per_40` · `college_last_rebounds_per_40` · `college_last_assists_per_40` · `college_last_steals_per_40` · `college_last_blocks_per_40` · `college_last_turnovers_per_40` · `college_last_field_goal_attempts_per_40` · `college_last_three_point_attempts_per_40` · `college_last_free_throw_attempts_per_40` · `college_last_assist_turnover_ratio` · `college_last_three_attempt_rate` · `college_last_free_throw_rate` · `college_last_ft_pct_shrunk` · `college_last_playmaking_margin_per_40` · `college_last_production_index` · `college_last_stocks_per_40_advanced` · `college_last_three_pct_shrunk` · `college_last_usage_proxy_per_40`

NCAA career · 32

`college_career_seasons` · `college_career_span_years` · `college_career_games_played` · `college_career_minutes` · `college_career_minutes_per_game` · `college_career_start_rate` · `college_career_field_goals_attempted` · `college_career_three_point_field_goals_attempted` · `college_career_free_throws_attempted` · `college_career_fg_pct` · `college_career_three_pct` · `college_career_ft_pct` · `college_career_efg_pct` · `college_career_ts_pct` · `college_career_points_per_40` · `college_career_rebounds_per_40` · `college_career_assists_per_40` · `college_career_steals_per_40` · `college_career_blocks_per_40` · `college_career_turnovers_per_40` · `college_career_field_goal_attempts_per_40` · `college_career_three_point_attempts_per_40` · `college_career_free_throw_attempts_per_40` · `college_career_assist_turnover_ratio` · `college_career_three_attempt_rate` · `college_career_free_throw_rate` · `college_career_ft_pct_shrunk` · `college_career_playmaking_margin_per_40` · `college_career_production_index` · `college_career_stocks_per_40_advanced` · `college_career_three_pct_shrunk` · `college_career_usage_proxy_per_40`

Development · 16

college_development_season_gap · college_development_minutes_per_game_delta · college_development_start_rate_delta · college_development_three_pct_delta · college_development_ft_pct_delta · college_development_efg_pct_delta · college_development_ts_pct_delta · college_development_points_per_40_delta · college_development_rebounds_per_40_delta · college_development_assists_per_40_delta · college_development_steals_per_40_delta · college_development_blocks_per_40_delta · college_development_turnovers_per_40_delta · college_development_assist_turnover_ratio_delta · college_development_three_attempt_rate_delta · college_development_free_throw_rate_delta

Age/upside and composite interactions · 6

college_age_on_july_1 · prospect_age_centered_21 · prospect_age_squared · prospect_youth_score · college_age_adjusted_scoring · college_age_adjusted_stocks

Categorical · 2

position · college_position

Advanced minutes-regression feature inventory

Base draft, age, and college features · 17

overall_pick · age_on_draft_year_july1 · age_on_draft_year_july1_missing · college_last_assists_per_40 · college_last_assist_turnover_ratio · college_last_points_per_40 · college_last_rebounds_per_40 · college_last_three_pct · college_last_ft_pct · college_last_stocks_per_40 · college_last_vs_career_points_per_40 · college_last_vs_career_steals_per_40 · college_last_vs_career_blocks_per_40 · college_last_vs_career_turnovers_per_40 · college_last_vs_career_three_pct · college_last_vs_career_ft_pct · college_last_vs_career_three_attempt_rate

Nonlinear draft capital · 8

draft_pick_log · draft_pick_sqrt · draft_pick_inverse · draft_pick_squared · draft_pick_top_5 · draft_pick_top_10 · draft_pick_lottery · draft_pick_first_round

Age, upside, and draft interaction · 6

prospect_age_centered_21 · prospect_age_squared · prospect_youth_score · college_age_adjusted_scoring · college_age_adjusted_stocks · draft_pick_x_age

Reliability and production composites · 8

college_career_ft_pct_shrunk · college_career_playmaking_margin_per_40 · college_career_production_index · college_career_stocks_per_40_advanced · college_career_three_pct_shrunk · college_last_playmaking_margin_per_40 · college_last_production_index · college_last_stocks_per_40_advanced

Last-season to career ratios · 9

college_last_to_career_assists_per_40_ratio · college_last_to_career_blocks_per_40_ratio · college_last_to_career_ft_pct_ratio · college_last_to_career_points_per_40_ratio · college_last_to_career_rebounds_per_40_ratio · college_last_to_career_steals_per_40_ratio · college_last_to_career_three_pct_ratio · college_last_to_career_ts_pct_ratio · college_last_to_career_turnovers_per_40_ratio

Reproduction commands and source map

Frozen training runs

```
python src/train_classifier.py --config configs/xgboost_draft_classifier_advanced_best.json
python src/train_model.py --config configs/xgboost_minutes_advanced_best.json
```

Classifier tuning

```
# quick check: five sampled candidates plus two configured fixed candidates
python src/tune_classifier.py --config configs/xgboost_draft_classifier_advanced_best.json -
-n-iter 5

# full recorded search: 180 sampled candidates plus two fixed candidates
python src/tune_classifier.py --config configs/xgboost_draft_classifier_advanced_best.json
```

Generated classifier artifacts

- `models/xgboost_draft_classifier_advanced.joblib` — estimator, sigmoid calibrator, decision rule, and schema.
- `models/xgboost_draft_classifier_advanced_metrics.json` — raw/calibrated metrics and purge metadata.
- `models/xgboost_draft_classifier_advanced_test_predictions.csv` — row-level holdout probabilities, decisions, and annual ranks.
- `models/xgboost_draft_classifier_advanced_feature_importance.csv` — fitted XGBoost split importance.

Implementation source map

Area	Primary source
Classifier command-line orchestration	<code>src/train_classifier.py</code>
Classifier preprocessing, calibration, decisions, and artifact helpers	<code>src/classifier_workflow.py</code>
Expanding-fold classification tuning and normalized AP	<code>src/tune_classifier.py</code> , <code>src/tuning.py</code>
Frozen classifier schema and parameters	<code>configs/xgboost_draft_classifier_advanced_best.json</code>
Regression preprocessing and training	<code>src/train_model.py</code> , <code>src/tune_model.py</code>
Frozen minutes schema and parameters	<code>configs/xgboost_minutes_advanced_best.json</code>
Shared two-stage table	<code>src/prepare_data.py</code>
Advanced feature formulas and materialization provenance	<code>.archive/src/advanced_feature_engineering.py</code> , <code>.archive/src/feature_engineering.py</code> , <code>.archive/src/build_advanced_model_datasets.py</code>
Classifier cohort and NCAA enrichment provenance	<code>nba_draft_data_collection/build_draft_classification_dataset.py</code> , <code>nba_draft_data_collection/build_enriched_draft_classification_dataset.py</code>
Historical verification reports	<code>.archive/models/xgboost_draft_classifier_advanced_tuning_results.json</code> , <code>.archive/models/xgboost_minutes_advanced_tuning_results.json</code>

Modeling limitations

- The classifier applies to Combine participants, not every NCAA player.
- The minutes model applies only to drafted players and includes draft capital; it is not a pre-draft talent-only model.
- Missing NCAA linkage is retained as missing. If linkage failure is systematically related to prospect type, native missing handling does not remove that selection mechanism.
- Multiple model families have now been compared on the same 2020–2022 period. Treat holdout values as descriptive and use a future cohort for the next independent confirmation.
- Predictions are predictive associations, not causal estimates of what a team or player should do.

Prepared from the checked-in code, configurations, materialized parquet schemas, and archived experiment reports in `D:\Mestrado\nba-draft`. Values were revalidated during the compatibility restoration on 19 August 2026.